TEMPORAC: IDENTITY-INDEXED LOCAL TEMPO ROUTING FOR MULTI-PERSON REPETITION COUNTING

Weiping Yan

University of Minnesota Twin Cities
yan00944@umn.edu

DRAFT – RESULTS PENDING

ABSTRACT

Multi-person repetition counting estimates a count for each supplied persistent identity track rather than only a scene total. The task is difficult when people move asynchronously, individual pace changes within a trajectory, and missing poses turn each trajectory into a masked nonstationary signal. We propose an identity-indexed signal-processing specification that receives supplied pose tracks and targets counting without person-wise count or period labels, while acknowledging external supervision in the pose and tracking frontends. Overlapping windows share an encoder and fast, medium, and slow response experts; masked autocorrelation and Fourier evidence yield local period cues and confidence. Soft routing mixes expert responses, and normalized overlap-add reconstructs one response and decodes it once per identity. This design separates per-track state from shared parameters and avoids independently counting every overlapping window, removing one structural source of boundary double-counting. Experimental evidence is pending synchronization with the collaborator's frozen implementation and audited MultiRep runs.

Index Terms— repetition counting, multi-person signals, local tempo, pose trajectories, temporal routing

1. INTRODUCTION

Person-wise repetition counts matter whenever several people exercise, rehabilitate, train, or perform repetitive work in the same scene. A single scene total cannot say who completed each repetition, paused, or changed pace. We therefore study multi-person repetition counting from supplied identity-indexed pose tracks. For N tracks, the primary output is a vector of N counts; their sum is only a diagnostic. The signal is difficult because people move asynchronously, a person's period may change inside one track, and occlusion creates irregular missing intervals. The primary claim is conditional on supplied persistent identity tracks; predicted-track experiments separately measure departures from that assumption.

Early work addressed live and real-world repetition counting [1, 2]; later temporal similarity and correlation models include RepNet and TransRAC [3, 4]. PoseRAC emphasizes salient poses [5], while PAMS learns period-adaptive multi-scale consistency from pose sequences without count labels in its counting objective [6]. MultiCounter establishes multi-person counting in untrimmed video [7]; MultiCounter+ further models spatiotemporal consistency and long-versus-short repetition regimes [8]. These advances cover multi-person outputs, pose-driven counting, low-label objectives, and broad period adaptation. They do not, however, resolve the narrower signal problem considered here: routing each overlapping window of each supplied identity track according to tempo that changes within that track, then reconstructing one response before count decoding. Recent counters also explore dynamic queries and decoupled spatiotemporal attention [9, 10]; those architectural directions are complementary to the identity-local temporal scale selection studied here. Windowed spectral estimation, frequency analysis under irregular sampling, short-time transform reconstruction, and learned expert routing are established ideas [11–14]; no one of them is claimed as novel here. Our narrower, result-free hypothesis is that identity-local tempo routing plus reconstruct-before-decode is more robust to within-track tempo changes than a parameter-matched global-tempo counter while preserving behavior on stationary tracks.

The proposed formulation separates inter-person asynchrony from intra-person nonstationarity. Each pose track is a masked multidimensional signal with identity-indexed state, whereas learned encoder and expert parameters are shared across people. The intended counting objective uses neither person-wise count labels nor person-wise period labels, but this qualifier does not extend to the perception stack: pose estimators and trackers may use externally supervised training data. The exact supervision path, including model selection and calibration, remains SYNC-REQUIRED: supervision firewall until the collaborator implementation is frozen. Selecting a global long-versus-short regime differs from allowing several pace changes inside one identity trajectory. Mask-aware local estimation also exposes missing observations instead of conflating invalid frames with stationary motion. These distinctions drive both the reconstruction invariant and the planned diagnostic.

Our signal path follows a fixed order. A shared encoder represents overlapping masked windows; local autocorrelation and Fourier evidence estimate period and confidence; a soft router mixes fast, medium, and slow shared response experts; normalized overlap-add reconstructs one response per identity; and one trajectory-level decoder processes it once. Figure 2 shows the conceptual specification. The local worktree does not yet contain this multi-person implementation, so the router, experts, reconstruction, decoder, and auxiliary objectives remain synchronization-required interfaces.

The pre-results manuscript makes three falsifiable contributions:

- It formulates person-wise counting as inference on identity-indexed, masked, nonstationary pose signals, with scene totals kept diagnostic.
- It specifies window-local tempo evidence, soft routing over shared response experts, normalized reconstruction, and one decode per track; every source-sensitive interface is explicitly marked for synchronization.
- It specifies planned protocol-separated comparisons, paired mechanism ablations, and a piecewise time-warp diagnostic. Empirical claims remain disabled until frozen MultiRep artifacts pass the evidence gates.

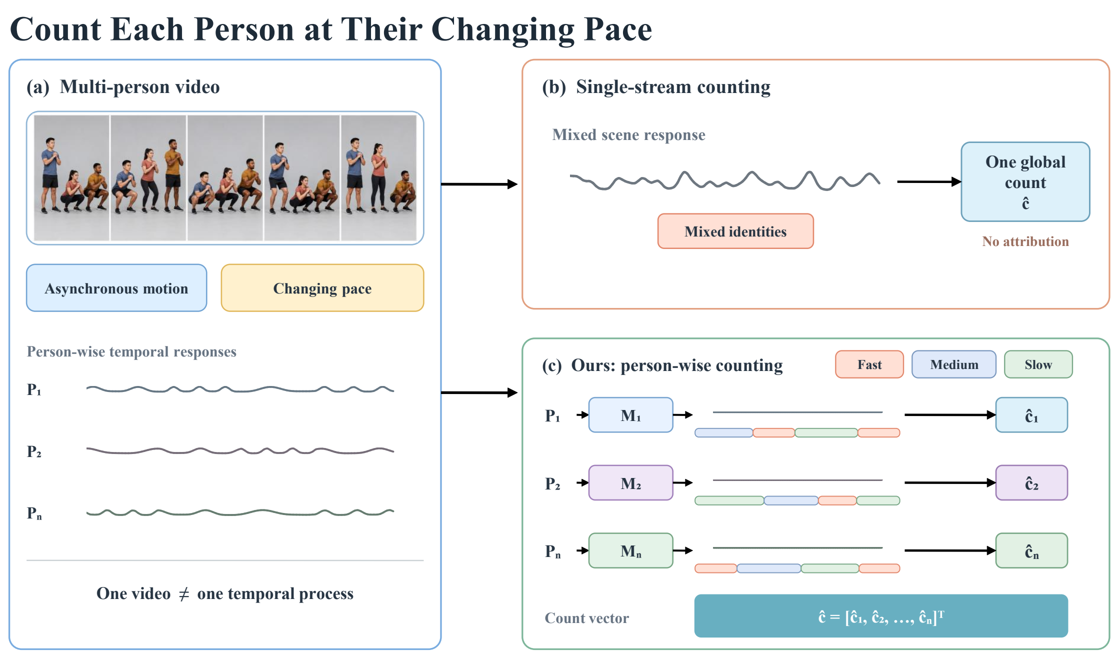

Fig. 1. Task motivation reproduced in full from the project's editable introduction deck. The drawing contrasts a mixed scene response with the desired person-wise responses under asynchronous motion and changing pace. Panel (b) is a naive scene-level adaptation rather than an attributed prior method. All curves and counts are schematic and are not measurements.

2. IDENTITY-INDEXED LOCAL TEMPO ROUTING

The counter begins after perception: it receives pose trajectories with supplied identity indices and validity masks. Oracle tracks, tracks from a frozen common frontend, and native method pipelines define different experimental protocols and are never ranked together. Parameters are shared, whereas buffers, periodic evidence, masks, and responses remain identity-indexed; no window combines people. Figure 2 summarizes the proposed specification, whose tensor layouts and learned modules remain SYNC-REQUIRED: collaborator source binding. For predicted tracks, an index denotes a current tracklet, not a guaranteed person. Fragmentation, re-entry, switches, retirement, and any reconciliation belong to the SYNC-REQUIRED: track lifecycle contract.

2.1. Masked identity-indexed representation

For person $i \in \{1, \ldots, N\}$, let $\mathbf{X}_i$ contain J joints with C coordinates over T_i frames, and let $m_i(t)$ mark a valid observation. Overlapping supports $\mathcal{W}_{i\ell}$ select local windows. A shared encoder $\mathcal{E}_\theta$ maps the masked samples to features, while the primary output remains a person-wise count vector:

$$\begin{gathered}\mathbf{X}_i \in \mathbb{R}^{T_i \times J \times C} \qquad m_i \in \mathbb{B}^{T_i} \qquad \widehat{\mathbf{c}} = [\hat{c}_i]_{i=1}^{N} \\ \mathbf{z}_{i\ell}(t) = \mathcal{E}_\theta(\mathbf{X}_i[\mathcal{W}_{i\ell}] \mid m_i[\mathcal{W}_{i\ell}])_t \qquad t \in \mathcal{W}_{i\ell}\end{gathered} \tag{1}$$

Here $\mathbb{B}$ denotes the binary alphabet. The shared θ does not mix identity state. Window geometry, pose normalization, padding, short tracks, and buffer lifecycle are SYNC-REQUIRED: track tensor and lifecycle contract. Masks accompany every transformation, and windows never cross identity timelines. A scene total $\sum_i \hat{c}_i$ is diagnostic. Permutation, isolated-mask, and short-track tests will check these invariants once the source interface exists.

2.2. Local masked periodic evidence

Tempo must vary with the window rather than being fixed for the whole track. Let $s_{i\ell}(t)$ denote the scalar evidence signal provisionally projected from $\mathbf{z}_{i\ell}(t)$, and $q_{i\ell}(t)$ its windowed validity mask. We define a mask-normalized non-circular autocorrelation and a normalized Fourier score, then map both to a period estimate and confidence:

$$\begin{aligned}\bar{s}_{i\ell} &= \frac{\sum_t q_{i\ell}(t) s_{i\ell}(t)}{\sum_t q_{i\ell}(t) + \epsilon} \\ A_{i\ell}(\tau) &= \frac{\sum_t q_{i\ell}(t) q_{i\ell}(t+\tau)[s_{i\ell}(t) - \bar{s}_{i\ell}][s_{i\ell}(t+\tau) - \bar{s}_{i\ell}]}{\sum_t q_{i\ell}(t) q_{i\ell}(t+\tau) + \epsilon} \\ F_{i\ell}(\omega) &= \frac{\left|\sum_t q_{i\ell}(t)[s_{i\ell}(t) - \bar{s}_{i\ell}] e^{-\mathrm{j}\omega t}\right|^2}{\max_{\omega'} \left|\sum_t q_{i\ell}(t)[s_{i\ell}(t) - \bar{s}_{i\ell}] e^{-\mathrm{j}\omega' t}\right|^2 + \epsilon} \\ \begin{bmatrix}\hat{p}_{i\ell} \\ \gamma_{i\ell}\end{bmatrix} &= \Phi(\{A_{i\ell}(\tau)\}_{\tau \in \mathcal{T}} \parallel \{F_{i\ell}(\omega)\}_{\omega \in \Omega})\end{aligned} \tag{2}$$

Indices (i, ℓ) expose within-track drift. Projection, centering, domains and units, interpolation, confidence, and fallback are SYNC-REQUIRED: period estimator; Eq. 2 fixes only the semantics of local masked evidence. Confidence must measure available evidence rather than a post-routing label, and low-support windows must be masked or use one frozen fallback in training and evaluation. Since periodic visibility can induce spectral peaks, the implementation must set out-of-range masks and minimum lag-pair support and pass periodic/nonperiodic missingness controls. It must also choose pair-dependent centering or another irregular-sample estimator [12].

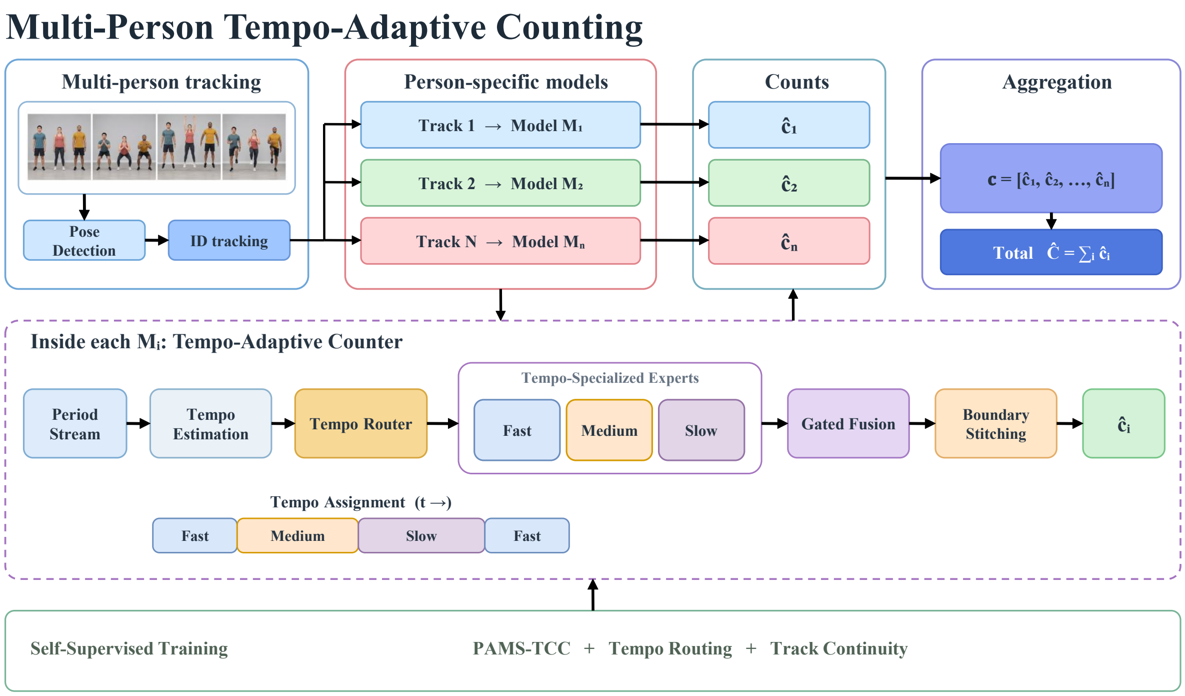

Fig. 2. Conceptual framework reproduced in full from the project's editable framework deck. In the manuscript, M_i denotes an identity-indexed counter instance with shared learned parameters and separate state, not a separately parameterized model. The training strip lists candidate objectives: tempo routing and track-continuity terms remain `SYNC-REQUIRED` until they are bound to frozen code. PAMS-TCC is attributed to PAMS [6]. The diagram is not implementation or result evidence.

2.3. Soft routing over shared response experts

Let $\mathcal{K} = \{\text{fast}, \text{medium}, \text{slow}\}$ index shared experts. A proposed router consumes the window representation, local period, and confidence, and returns simplex weights. Each expert emits a continuous local response rather than an independently rounded count:

$$
\begin{aligned}
\mathbf{u}_{i\ell} &= \rho_\psi([\mathbf{z}_{i\ell} \,\|\, \hat{p}_{i\ell} \,\|\, \gamma_{i\ell}]) \\
\mathbf{g}_{i\ell} &= \operatorname{softmax}(\mathbf{u}_{i\ell}/\tau_{\mathrm{r}}) \\
g_{i\ell k} &\geq 0 \qquad \sum_{k \in \mathcal{K}} g_{i\ell k} = 1 \\
\mathbf{r}^{(k)}_{i\ell} &= \mathcal{H}^{(k)}_{\phi_k}(\mathbf{z}_{i\ell}) \qquad k \in \mathcal{K}
\end{aligned}
\tag{3}
$$

Soft weights retain uncertainty at tempo transitions. Router inputs, temperature, expert architecture, gradients, sharing, and fallback remain SYNC-REQUIRED: router and expert implementation. Fast/medium/slow are intended scales, not observed specializations: paired controls must test expert use, collapse, frozen tempo bands, and hard/uniform routing. Shared experts keep parameter count independent of the number of tracks, but no runtime or memory claim is enabled without profiling.

2.4. Response reconstruction and one track decode

The final stage mixes expert responses before merging overlapping windows. Let $a_{i\ell}(t)$ be a taper on window ℓ, and let $q_{i\ell}(t)$ exclude invalid samples. Normalized overlap-add forms a single track response R_i, after which one decoder extracts events:

$$
\begin{aligned}
\widetilde{r}_{i\ell}(t) &= \sum_{k \in \mathcal{K}} g_{i\ell k}\, r^{(k)}_{i\ell}(t) \\
R_i(t) &= \frac{\sum_{\ell}_{t \in \mathcal{W}_{i\ell}} a_{i\ell}(t) q_{i\ell}(t) \widetilde{r}_{i\ell}(t)}{\sum_{\ell}_{t \in \mathcal{W}_{i\ell}} a_{i\ell}(t) q_{i\ell}(t) + \epsilon} \\
\widehat{\mathcal{E}}_i &= \mathcal{D}(R_i \mid m_i) \qquad \hat{c}_i = |\widehat{\mathcal{E}}_i|
\end{aligned}
\tag{4}
$$

This order avoids counting every window independently; it does not guarantee one peak. Decoded samples require aligned timestamps and coverage $\sum_\ell a_{i\ell}(t) q_{i\ell}(t) > 0$. Taper, gaps, event confidence, peak parameters, and decoder semantics remain SYNC-REQUIRED: reconstruction and decoder. PAMS-TCC is the documented starting objective [6]; $\mathcal{L}_{\text{PAMS-TCC}} + \lambda_{\mathrm{r}} \mathcal{L}_{\text{route}} + \lambda_{\mathrm{t}} \mathcal{L}_{\text{track}}$ is only a synchronization contract. The latter terms, their existence, formulas, weights, samples, supervision, and gradient paths must bind to code; absent terms will be deleted. Overlap-add preserves continuous evidence before one masked trajectory-level decode. A boundary test will place an event at several offsets and test one-time decoding.

The supervision audit must inspect loaders, pseudo-targets, auxiliary losses, selection, stopping, calibration, and inference fallbacks at one commit. Until it passes, the paper states only the intended absence of person-wise count and period labels in the counting objective and discloses perception supervision.

Table 1. Protocol-separated main comparison. Arrows indicate metric direction; pending cells contain no measurement.

Method	Protocol	P-mAP ↑	AP50 ↑	AP75 ↑	MAE ↓	OBO ↑
MultiCounter [7]	native	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+ [8]	native	PENDING	PENDING	PENDING	PENDING	PENDING
Track-PAMS [6]	oracle	PENDING	PENDING	PENDING	PENDING	PENDING
Global tempo	oracle	PENDING	PENDING	PENDING	PENDING	PENDING
TempoRAC	oracle	PENDING	PENDING	PENDING	PENDING	PENDING
Track-PAMS [6]	common track	PENDING	PENDING	PENDING	PENDING	PENDING
Global tempo	common track	PENDING	PENDING	PENDING	PENDING	PENDING
TempoRAC	common track	PENDING	PENDING	PENDING	PENDING	PENDING

Table 2. Paired mechanism ablations. Deltas are variant minus the soft-routing NOLA reference.

Tempo	Cue	Route	Merge	ΔMAE	ΔOBO
global	A+F	uniform	NOLA	PENDING	PENDING
local	ACF	soft	NOLA	PENDING	PENDING
local	FFT	soft	NOLA	PENDING	PENDING
local	A+F	single	NOLA	PENDING	PENDING
local	A+F	hard	NOLA	PENDING	PENDING
local	A+F	soft	boxcar	PENDING	PENDING
local	A+F	soft	window	PENDING	PENDING
local	A+F	soft	NOLA	0	0

3. PLANNED EXPERIMENTS AND ANALYSIS

The protocol is prospectively specified here but not yet executed for the proposed system. Existing single-person development artifacts neither implement the method nor use the target benchmark and are excluded. Result prose is enabled only after a frozen collaborator bundle passes experiment and claim audits.

3.1. Protocol and metrics

The manifest must bind the MultiRep release, official split, media and annotation checksums, evaluator revision, preprocessing, supervision path, hardware, configurations, seeds, and raw logs. We report three noninterchangeable protocols: supplied oracle tracks, one frozen common predicted-track frontend, and each method's native pipeline. Native rows provide context and are not ranked against common-track rows. Predicted-track runs retain HOTA [15], IDF1, and identity switches [16] as frontend diagnostics, separate from counting. Their proposed-method values remain PENDING, PENDING, and PENDING.

The principal release-native metrics are Period-mAP, Period-AP50, Period-AP75, AvgMAE, and AvgOBO [7, 8]. The manifest freezes the evaluation unit, assignment and unmatched-track rules, aggregation, empty-support handling, and evaluator hash. Reproduced learned methods use at least three predeclared seeds; we report mean, sample standard deviation, and a paired video-cluster bootstrap 95% confidence interval. The dataset and evaluator remain PENDING and PENDING until their hashes are available.

3.2. Matched comparisons and ablations

Table 1 separates native MultiCounter variants from matched oracle- and common-track families. Oracle tracks isolate counting; the common frontend exposes tracking errors. Track-PAMS tests per-track adaptation, a global-tempo control tests locality, and TempoRAC tests the full hypothesis. RepNet, TransRAC [3, 4], PoseRAC [5], DeTRC [9], and D^2-STX [10] enter a matched block only when input, output, supervision, and evaluation parity can be established; otherwise they remain contextual.

Table 2 isolates local versus global tempo, ACF and Fourier evidence, uniform/hard/soft or single-expert routing, and boundary fusion. The window variant decodes each window before one frozen reconciliation rule. Every row must match parameters, context, frontend, training budget, videos, and seeds, and must bind one commit and configuration. Auxiliary-loss removals are added only for terms present in the frozen graph. Primary contrasts, bootstrap replicates, multiplicity, and exclusion rules are fixed before inspection.

3.3. Nonstationary-tempo diagnostic

A frozen piecewise time-warp changes one person's pace while preserving paired identity and event count. The result-bound diagnostic aligns local period, routing probability or entropy, reconstructed response, and clean-to-corrupt change, using either a frozen selection rule or a predeclared aggregate. Source data, generator hash, warp schedule, frame/mask/event mapping, and selection rule are recorded. Separate controls hold motion fixed under periodic and nonperiodic missingness, then hold the mask fixed while changing tempo, testing whether routing reads motion rather than visibility. Occlusion and identity switches are stressed separately so that perception failures are not presented as tempo effects.

4. CONCLUSION

We formulated multi-person repetition counting as inference on masked, identity-indexed pose signals whose tempo can change within a track. The proposed specification estimates local periodic evidence, softly routes shared response experts, reconstructs each identity response with normalized overlap-add, and decodes that response once. This pre-results draft has no local multi-person implementation or eligible MultiRep results, and its pose and tracking frontends rely on external supervision. Completion requires a frozen collaborator commit, source-bound losses and interfaces, matched oracle and common-track evaluation, and paired piecewise-tempo diagnostics. Empirical and citation audits must then be rerun before any quantitative or comparative conclusion is enabled.

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